

STATISTICS II

- Q1** Consider two random variables X and Y with joint PMF given in Table.

	Y=2	Y=4	Y=5
X=1	1/12	1/24	1/24
X=2	1/6	1/12	1/8
X=3	1/4	1/8	1/12

- Find $P(X \leq 2, Y \leq 4)$.
 - Find the marginal PMF's of X and Y.
 - Find $P(Y=2 | X=1)$.
 - Are X and Y independent?
- Q2** For the adjoining bivariate probability distribution of X and Y, find:
- $P(X \leq 1, Y = 2)$,
 - $P(X \leq 1)$,
 - $P(Y \leq 3)$, and
 - $P(X < 3, Y \leq 4)$.

y \ x	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

- Q3** For the joint probability distribution of two random variable X and Y given below:

Y \ X	1	2	3	4	Total
1	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{10}{36}$
2	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{9}{36}$
3	$\frac{5}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{8}{36}$
4	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{5}{36}$	$\frac{9}{36}$
Total	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	1

Find 1) the marginal distributions of X and Y, and
2) conditional distribution of X given the value of $Y=1$ and that of Y given the value of $X=2$.

- Q4** I have a bag containing 40 blue marbles and 60 red marbles. I choose 10 marbles (without replacement) at random. Let X be the number of blue marbles and y be the number of red marbles. Find the joint PMF of X and Y.

- Q5** The joint p.d.f of two random variable X and Y is given by :

$$f(x, y) = \frac{9(1+x+y)}{2(1+x)^4(1+y)^4}; \begin{cases} 0 \leq x < \infty \\ 0 \leq y < \infty \end{cases}$$

Find the marginal distribution of X and Y, and the conditional distribution of Y for $X=x$.

- Q6** Given:

$$f(x, y) = e^{-(x+y)}; 0 \leq x < \infty, 0 \leq y < \infty$$

find

- $P(X > 1)$
- $P(X < Y | X < 2Y)$
- $P(1 < X + Y < 2)$

- Q7** Let X and Y be jointly continuous random variables with Joint PDF

$$f_{X,Y}(x, y) = \begin{cases} cx + 1 & x, y \geq 0, x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c.
- Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
- Find $P(Y < 2X^2)$.

- Q8** Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6e^{-(2x+3y)} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Are X and Y independent?
- Find $E[Y|X > 2]$.
- Find $P(X > Y)$.

- Q9** Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



We know that given $X = x$, the random variable Y is uniformly distributed on $[-x, x]$.

- 1) Find the joint PDF $f_{XY}(x, y)$.
- 2) Find $f_Y(y)$.

Q10 A bank operates a drive-up and a walk-up window. Here

X = the proportion of time the drive-up facility is in use.

Y = the proportion of time the walk-up window is in use.

$f(x, y)$

$$= \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

The probability that neither facility is busy more than one-quarter of the time is?



Answer Key

Q1	$\frac{3}{8}, 1/2$	Q8	1) $f_X(x) = 2e^{-2x}, x \geq 0$ and $f_Y(y)$ $= 3e^{-3y}, y \geq 0$	
Q2	1/16, 28/32, 23/64, 9/16		2) $\frac{1}{3}$	
Q3	Refer to hint for solution		3) $\frac{3}{5}$	
Q4	Refer to hint for sol			
Q5	Refer to the hint for solution	Q9	$f_{XY}(x, y) \begin{cases} 1 \\ 0 \end{cases}$	$ y \leq x \leq 1$ <i>otherwise</i>
Q6	$e^{-1}, \frac{3}{4}, \frac{2}{e} - \frac{3}{e^2}$	Q10	7/640	
Q7	3, 53/96			



Hints & Solutions

Q1 Text Solution:

1)

To find $P(X \leq 2, Y \leq 4)$, we can write

$$P(X \leq 2, Y \leq 4) = P_{XY}(1, 2) + P_{XY}(1, 4) + P_{XY}(2, 2) + P_{XY}(2, 4)$$

$$= \frac{1}{12} + \frac{1}{24} + \frac{1}{6} + \frac{1}{12} = \frac{3}{8}.$$

2)

Note from the table that

 $R_X = \{1, 2, 3\}$ and $R_Y = \{2, 4, 5\}$

$$P_X(y) = \begin{cases} \frac{1}{2} & y = 2 \\ \frac{1}{4} & y = 4 \\ \frac{1}{4} & y = 5 \\ 0 & \text{otherwise} \end{cases}$$

$$P_X(x) = \begin{cases} \frac{1}{6} & x = 1 \\ \frac{3}{8} & x = 2 \\ \frac{11}{12} & x = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$P_X(y) = \begin{cases} \frac{1}{2} & y = 2 \\ \frac{1}{4} & y = 4 \\ \frac{1}{4} & x = 5 \end{cases}$$

3) Using the formula for conditional probability, we have

$$P(Y = 2|X = 1) = \frac{P(X=1 \cap Y=2)}{P(X=1)}$$

$$= \frac{P_{XY}(1,2)}{P_X(1)}$$

$$= \frac{\frac{1}{12}}{\frac{1}{6}} = \frac{1}{2}$$

4)

 $P(X=x_i \cap Y=y_j)$ is equal to $P(X=x_i) \cdot P(Y=y_j)$

$$P(X = 2, Y = 2) = \frac{1}{6}$$

$$\neq P(X = 2)P(Y = 2) = \frac{3}{16}.$$

Thus, we conclude that X and Y are not independent.

Q2 Text Solution:

The marginal distributions are given below:

	y	1	2	3	4	5	6	$P_X(x)$
x								

0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{8}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{10}{16}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$	$\frac{8}{64}$
$P_Y(y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	$\Sigma P(x) = 1$ $\Sigma P(y) = 1$

$$(i) P(X \leq 1, Y = 2) = P(X = 0, Y = 2) + P(X = 1, Y = 2) = 0 + \frac{1}{16} = \frac{1}{16}$$

$$(ii) P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{8}{32} + \frac{10}{16} = \frac{7}{8}$$

$$(iii) P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

$$(iv) P(X < 3, Y \leq 4) = P(X = 0, Y \leq 4) + P(X = 1, Y \leq 4) + P(X = 2, Y \leq 4)$$

$$= \left(\frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right) = \frac{9}{16}$$

Q3 Text Solution:

(i) The marginal distributions of X is defined as:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$P(X = 1) = \sum_y P(X = 1, Y = y)$$

$$= P(X = 1, Y = 1) + P(X = 1, Y = 2) + P(X = 1, Y = 3) + P(X = 1, Y = 4)$$

$$= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$$

Similarly,

$$P(X = 2) = \sum_y P(X = 2, Y = y) = \frac{9}{36};$$

$$P(X = 3) = \sum_y P(X = 3, Y = y) = \frac{8}{36}$$

$$P(X = 4) = \sum_y P(X = 4, Y = y) = \frac{9}{36}.$$

Similarly, we can obtain the marginal distribution of Y .

Value of X, x	1	2	3	4
$P(X = x)$	$\frac{1}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

Value of Y, y	1	2	3	4
$P(Y = y)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

(ii) The conditional probability function of X given Y is defined as follows:

$$P(X = x|Y = y) = \frac{P(X=x|Y=y)}{P(Y=y)}.$$
 Therefore

$$P(X = 1|Y = 1) = \frac{P(X=1|Y=1)}{P(Y=1)} = \frac{4/36}{11/36} = \frac{4}{11}$$


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$$\begin{aligned}
 P(X=2|Y=1) &= \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} \\
 &= \frac{1}{11} \\
 P(X=3|Y=1) &= \frac{P(X=3, Y=1)}{P(Y=1)} = \frac{5/36}{11/36} \\
 &= \frac{5}{11} \\
 P(X=4|Y=1) &= \frac{P(X=4, Y=1)}{P(Y=1)} = \frac{1/36}{11/36} \\
 &= \frac{1}{11}
 \end{aligned}$$

Hence the conditional distribution of X given Y = 1 is:

x :	1	2	3	4
$P(X=x Y=1)$	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{4}{11}$

Similarly, we can obtain the condition of Y for X = 2 as given below:

y :	1	2	3	4
$P(Y=y X=2)$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{2}{9}$

Q4 Text Solution:

This is, in fact a hypergeometric distribution. First note that we must have $X+Y=10$, So

$$\begin{aligned}
 P_{XY} &= \{(i, j) | i+j=10, i, j \in \mathbb{Z}, i, j \geq 0\} \\
 &= \{(0, 10), (1, 9), (2, 8), \dots, (10, 0)\}.
 \end{aligned}$$

$$P_{XY}(i, j) = \begin{cases} \frac{\binom{40}{i} \binom{60}{j}}{\binom{100}{10}} & i+j=10, i, j \in \mathbb{Z}, i, j \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Q5 Text Solution:

Marginal p.d.f of X is given by

$$\begin{aligned}
 f_x(x) &= \int_0^\infty f(x, y) dy \\
 &= \frac{9}{2(1+x)^4} \cdot \int_0^\infty \frac{(1+y)^x}{(1+y)^4} dy \\
 &= \frac{9}{2(1+x)^4} \cdot \int_0^\infty \left[(1+y)^{-3} + x(1+y)^{-4} \right] dy \\
 &= \frac{9}{2(1+x)^4} \cdot \left[\left| \frac{-1}{2(1+y)^2} \right|_0^\infty + x \left| \frac{-1}{3(1+y)^3} \right|_0^\infty \right] \\
 &= \frac{9}{2(1+x)^4} \cdot \left[\frac{1}{2} + \frac{x}{3} \right] \\
 &= \frac{3}{4} \cdot \frac{3+2x}{(1+x)^4}; 0 \leq x < \infty
 \end{aligned}$$

Since $f(x, y)$ is symmetric in x and y, the marginal p.d.f of y is given by

$$\begin{aligned}
 f_y(y) &= \int_0^\infty f(x, y) dx \\
 &= \frac{3}{4} \cdot \frac{3+2y}{(1+y)^4}; 0 \leq y < \infty
 \end{aligned}$$

The conditional distribution of Y for $X=x$ is given by

$$\begin{aligned}
 f_{xy}(Y=y|X=x) &= \frac{f_{xy}(x, y)}{f_x(x)} \\
 &= \frac{9(1+x+y)}{2(1+x)^4(1+y)^4} \cdot \frac{4(1+x)^4}{3(3+2x)} \\
 &= \frac{6(1+x+y)}{(1+y)^4(3+2x)}; 0 \leq y < \infty
 \end{aligned}$$

Q6 Text Solution:

We are given:

$$f(x, y) = e^{-(x+y)}; 0 \leq x < \infty, 0 \leq y < \infty$$

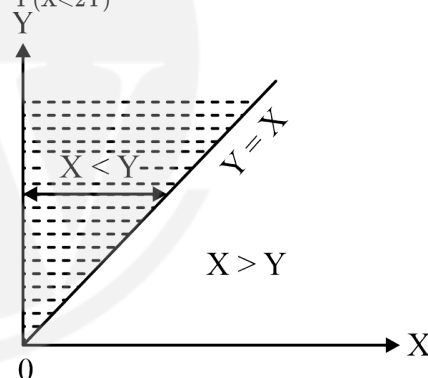
$$\begin{aligned}
 &\dots(1) \\
 &= (e^{-x})(e^{-y}) \\
 &= f_x(x) \cdot f_y(y); 0 \leq x < \infty, 0 \leq y < \infty \\
 &\Rightarrow X \text{ and } Y \text{ are independent and} \\
 &f_x(x) = e^{-x}; x \geq 0 \text{ and } f_y(y) = e^{-y}; y \geq 0 \\
 &\dots(2)
 \end{aligned}$$

$$1) P(X > 1) = \int_1^\infty f_x(x) dx = \int_1^\infty e^{-x} dx$$

$$= \left| \frac{e^{-x}}{-1} \right|_1^\infty = \frac{1}{e}$$

$$2) P(X < Y | X < 2Y) = \frac{P(X < Y \cap X < 2Y)}{P(X < 2Y)}$$

$$\frac{P(X < Y)}{P(X < 2Y)} \dots(3)$$



$$\begin{aligned}
 P(X < Y) &= \int_0^\infty \left[\int_0^y f(x, y) dx \right] dy \\
 &= \int_0^\infty \left[e^{-y} \left| \frac{e^{-x}}{-1} \right|_0^y \right] dy = - \int_0^\infty e^{-y} (e^{-y} - 1) dy \\
 &= \left| \frac{e^{-2y}}{-2} + e^{-y} \right|_0^\infty = 1 - \frac{1}{2} = \frac{1}{2} \\
 P(X < 2Y) &= \int_0^\infty \left[\int_0^{2y} f(x, y) dx \right] dy = - \\
 &\int_0^\infty e^{-y} (e^{-2y} - 1) dy
 \end{aligned}$$

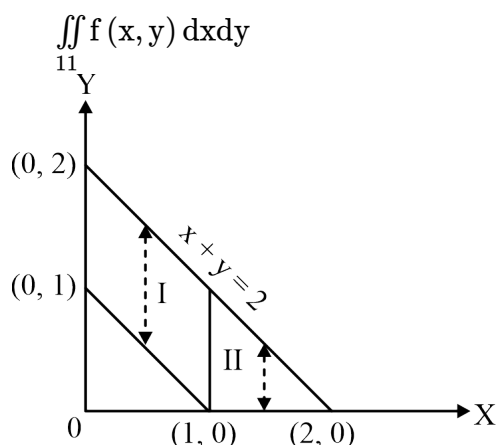
$$= - \left| \frac{e^{-3y}}{-3} + e^{-y} \right|_0^\infty = 1 - \frac{1}{3} = \frac{2}{3}$$

Substituting in (3)

$$P(X < Y | X < 2Y) = \frac{1/2}{2/3} = \frac{3}{4}$$



$$3) P(1 < X + Y < 2) = \iint_1 f(x, y) dx dy =$$



Random Variables-Distribution Functions

$$\begin{aligned}
 &= \int_0^1 \left[\int_{1-x}^{2-x} f(x, y) dy \right] dx + \\
 &\int_1^2 \left[\int_0^{2-x} f(x, y) dy \right] dx \\
 &= \int_0^1 \left[e^{-x} \int_{1-x}^{2-x} e^{-y} dy \right] dx + \\
 &\int_1^2 \left[e^{-x} \int_0^{2-x} e^{-y} dy \right] dx \\
 &= \int_0^1 \left[\frac{e^{-x}}{-1} (e^{x-2} - e^{x-1}) \right] dx + \\
 &\int_1^2 \left[\frac{e^{-x}}{-1} (e^{x-2} - 1) \right] dx \\
 &= -(e^{-2} - e^{-1}) \int_0^1 1 \cdot dx - \int_1^2 (e^{-2} - e^{-x}) dx \\
 &= -(e^{-2} - e^{-1}) \left[x \right]_0^1 - \left[e^{-2} \cdot x + e^{-x} \right]_1^2 \\
 &= \frac{2}{e} - \frac{3}{e^2}
 \end{aligned}$$

Q7 Text Solution:

To find the constant c , we write

$$\begin{aligned}
 1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy \\
 &= \int_0^1 \int_0^{1-x} cx + 1 dy dx \\
 &= \int_0^1 (cx + 1)(1 - x) dx \\
 &= \frac{1}{2} + \frac{1}{6}c.
 \end{aligned}$$

Thus, we conclude $c = 3$.

2) We first note that $R_X = R_Y = [0, 1]$

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dy \\
 &= \int_0^{1-x} 3x + 1 dy \\
 &= (3x + 1)(1 - x), \text{ for } x \in [0, 1]
 \end{aligned}$$

Thus, we have

$$\begin{aligned}
 f_X(x) &= \begin{cases} (3x + 1)(1 - x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Similarly, we obtain

$$\begin{aligned}
 f_Y(y) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dx \\
 &= \int_0^{1-y} 3x + 1 dx \\
 &= \frac{1}{2}(1 - y)(5 - 3y), \text{ for } y \in [0, 1].
 \end{aligned}$$

Thus, we have

$$\begin{aligned}
 f_Y(y) &= \begin{cases} \frac{1}{2}(1 - y)(5 - 3y) & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

3) To find $P(Y < 2X^2)$ we need to integrate $f_{XY}(x, y)$ over the region shown in figure.

$$\begin{aligned}
 P(Y < 2X^2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dy dx \\
 &= \int_0^1 \int_0^{\min(2x^2, 1-x)} 3x + 1 dy dx \\
 &= \int_0^1 (3x + 1) \min(2x^2, 1 - x) dx \\
 &= \int_0^{\frac{1}{2}} (3x + 1) 2x dx + \\
 &\int_{\frac{1}{2}}^1 (3x + 1) (1 - x) dx \\
 &= 53/96
 \end{aligned}$$

Q8 Text Solution:

1) In this question we have-

$$f_{X,Y}(x, y) = f_X(x) f_Y(y),$$

Where

$$f_X(x) = 2e^{-2x}, x \geq 0$$

$$f_Y(y) = 3e^{-3y}, y \geq 0$$

Thus they are independent.

2) Since X and Y are independent we have

$$E[Y|X > 2] = E[Y]. \quad \text{Note that } Y \sim \text{Exponential}(3), \text{ thus } E(Y) = \frac{1}{3}.$$

3)

$$\begin{aligned}
 P(X > Y) &= \int_0^{\infty} \int_y^{\infty} 6e^{-(2x+3y)} dx dy \\
 &= \int_0^{\infty} 3e^{-5y} dy \\
 &= 3/5
 \end{aligned}$$



Q9 Text Solution:

1)

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{2x} & -x \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

Thus, we have

$$f(x, y) = \begin{cases} 1 & 0 \leq x \leq 1, -x \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

$$f(x, y) = \begin{cases} 1 & |y| \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

2)

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

$$= \int_{|y|}^1 1 dx$$

$$= 1 - |y|.$$

Thus,

$$f_Y(y) = \begin{cases} 1 - |y| & |y| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} p\left(0 < x < \frac{1}{4}, 0 < y < \frac{1}{4}\right) &= \int_0^{\frac{1}{4}} \int_0^{\frac{1}{4}} f(x, y) dx dy \\ &= \int_0^{\frac{1}{4}} \int_0^{\frac{1}{4}} \frac{6}{5} \left(x + y^2\right) dx dy \\ &= \frac{6}{5} \left(\int_0^{\frac{1}{4}} \left(\frac{x^2}{2} + xy^2\right) dy \right) \\ &= \frac{6}{20} \left(\frac{x^2}{2} + \frac{y^3}{3}\right)_0^{\frac{1}{4}} = \frac{7}{640} \end{aligned}$$

Q10 Text Solution:

$$f(x, y)$$

$$= \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Say the manager has given us the joint pdf based on his experience:

